ON 'SUB-GAMMA' / 'BERNSTEIN-TYPE' CONCENTRATION BOUNDS

Abstract. In the study of concentration inequalities, there is a popular notion of weaker-than-Gaussian concentration known variously as 'sub-Gamma' or 'Bernstein-type' bounds. In this note, we i) prove that any random variable admitting an absolute exponential moment will satisfy such a bound, and ii) that the bound is exactly saturated by a specific random variable of Compound Poisson type.

1. Introduction

Cumulant-generating functions (CGFs) play a central role in the study of large deviations theory and the concentration of measure phenomenon, with countless applications. While large deviations often concerns itself with exact characterisations of the CGF, for concrete applications, it often suffices to have a reasonably tractable upper bound. As a result, there exist several useful families of random variables which are characterised by their satisfaction of a suitable bound on their CGF.

An archetypal such instance is the family of sub-Gaussian random variables, perhaps discussed earliest by Kahane [3, 5], and characterised by the fact that their CGF is uniformly dominated by the CGF of a Gaussian random variable of suitable variance, i.e.

$$X \in \mathsf{sG}(v) \iff \forall t \in \mathbf{R}, \qquad \log\left(\mathbf{E}\left[\exp\left(t \cdot X\right)\right]\right) \leq v \cdot \frac{t^2}{2}.$$

This then immediately implies tail bounds for X which decay essentially as rapidly as for the centered Gaussian $\mathcal{N}(0, v)$ itself, i.e.

$$r \geq 0 \implies \max\left\{\mathbf{P}\left(X \geq r\right), \mathbf{P}\left(-X \geq r\right)\right\} \leq \exp\left(-\frac{r^2}{2 \cdot v}\right).$$

These bounds tend to be rather useful in applications. Moreover, observe that even for random variables with lighter-than-Gaussian tails, when interest centers upon high-dimensional models (and e.g. sums of large quantities of weakly-correlated variables), the Central Limit Theorem tends to imply that 'better-than-Gaussian' behaviour cannot really persist. In this regard, sub-Gaussian estimates are something of a gold standard, being essentially the strongest form of concentration which interfaces favourably with tensorisation.

Development of the CGF around $t = 0$ shows that when $X \in \mathsf{sG}(v)$, it necessarily holds that $\mathbf{E}[X] = 0$ and $\mathbf{E}\left[X^2\right] \leq v$, and so the tail bounds obtained in this way cannot be better than those for the centered Gaussian of matching variance. When the latter bound is saturated, i.e. $\mathbf{E}\left[X^2\right] = v$, then one says that X is strictly sub-Gaussian (see e.g. [1]), and the tail bounds so obtained are typically highly accurate. In particular, when studying iid sums of realisations of X, the implied tail bounds correctly reproduce the bounds implied by the Central Limit Theorem, up to sub-dominant terms.

A limitation of sub-Gaussian estimates is that they can nevertheless be somewhat loose in certain circumstances. While the class of strictly sub-Gaussian random variables is surprisingly broad in terms including various probabilistic familes of practical interest, it is structurally rather rigid. Indeed, it can be the case that $\mathbf{E}[X] = 0$, $\mathbf{E}\left[X^2\right] = 1$, and $\log\left(\mathbf{E}\left[\exp\left(t \cdot X\right)\right]\right) \in \mathcal{O}\left(t^2\right)$ globally, but that $X \in \mathsf{sG}(v)$ only for some $v \gg 1$. At an intuitive level, this reflects that certain random variables require some 'warm-up' before the asymptotic regime of the CLT becomes relevant. As an example, this arises when studying 'skewed' Bernoulli random variables, i.e.

$$X = \frac{\mathsf{Bernoulli}(p) - p}{\sqrt{p \cdot (1 - p)}}, \qquad p \approx 0,$$

for which their sum has to first 'look Poisson' before it can begin to 'look Gaussian'. In any case, in such settings, the tail bounds implied by the sub-Gaussian estimates are instead quite loose, failing to capture the essence of the Central Limit Theorem in a crucial way.

To address this situation, various refined notions of concentration have been proposed, typically involving multiple parameters to 'pin down when the CLT kicks in'. A popular example is the sub-Exponential class,

often defined by

$$X \in \mathsf{sE}(v,h) \iff \forall t \in [-h,h], \qquad \log\left(\mathbf{E}\left[\exp\left(t \cdot X\right)\right]\right) \leq v \cdot \frac{t^2}{2}.$$

This remains relatively easy to convert into tail bounds, as

$$r \geq 0 \implies \max\left\{\mathbf{P}\left(X \geq r\right), \mathbf{P}\left(-X \geq r\right)\right\} \leq \exp\left(-\min\left\{\frac{r^2}{2 \cdot v}, r \cdot h - \frac{v}{2} \cdot h^2\right\}\right).$$

Additionally, by trading off the value of h, one can obtain any $v > \mathbf{E}\left[X^2\right]$, allowing for almost-correct small-deviation asymptotics. There is arguably a slight conceptual wrinkle in that this bound does not correspond to any single random variable (i.e. there is no random variable X such that $\log\left(\mathbf{E}\left[\exp\left(t \cdot X\right)\right]\right) \leq v \cdot \frac{t^2}{2}$ for all $t \in [-h,h]$, and is otherwise infinite), but this point is admittedly of a rather philosophical nature.

Another popular class of a similar flavour is the 'sub-Gamma' [2] or 'Bernstein-type' [6] class, defined by

$$X \in \beta(v,b) \iff \forall t \in \left[-b^{-1}, b^{-1}\right], \qquad \log\left(\mathbf{E}\left[\exp\left(t \cdot X\right)\right]\right) \leq \frac{v \cdot t^2}{2 \cdot \left(1 - b \cdot |t|\right)},$$

and implying the tail bounds

$$r \geq 0 \implies \max\left\{\mathbf{P}\left(X \geq r\right), \mathbf{P}\left(-X \geq r\right)\right\} \leq \exp\left(-\frac{\left[\sqrt{v + 2 \cdot b \cdot r} - \sqrt{v}\right]^2}{2 \cdot b^2}\right) \leq \exp\left(-\frac{r^2}{2 \cdot \left(v + b \cdot r\right)}\right).$$

Observe that despite its name, this is **not** an upper bound involving the CGF of any Gamma random variable. Actually, if one restricts attention to $t \in \left[0, b^{-1}\right]$, we will later demonstrate that it is indeed the CGF of a rather different random variable. The saturating X can be defined in terms of a Compound Poisson random variable; this is not difficult to show, but seems not to have been remarked upon explicitly.

Moreover, by trading off b, one can always take $v = \mathbf{E}\left[X^2\right]$; we will show this shortly, also providing an explicit estimate for b. In this regard, the Bernstein bound is quite satisfactory in terms of i) reflecting a genuine comparison of CGFs, and ii) giving sharp CLT-type tail bounds at sufficiently-large sample size. The purpose of this note is to provide the details of these results.

2. Finite Absolute Exponential Moments imply a Bernstein Bound

The definition of Bernstein-type bounds implies directly that the moment- (or equivalently, cumulant-) generating function of X exists in some neighbourhood of $t \sim 0$. In this section, we establish that the converse holds true, i.e. if the moment-generating function of X exists in some neighbourhood of $t \sim 0$, then a Bernstein-type bound can be deduced. We establish this by making a specific quantitative assumption on the moment-generating function; one could easily make alternative assumptions and deduce similar conclusions.

Proposition 1. *Let X be a centered real-valued random variable of positive variance v, and let $h > 0$, $M > 0$ be such that*

$$\max\left\{\mathbf{E}\left[\exp\left(h \cdot X\right)\right], \mathbf{E}\left[\exp\left(-h \cdot X\right)\right]\right\} \leq M.$$

Writing $b_\star = \frac{4 \cdot (M-1) - v \cdot h^2}{v \cdot h^3}$, there then exists $b \in [0, b_\star]$ such that $X \in \beta(v,b)$.

Proof. Recall from e.g. Equation 2.15 of [6] that if there hold the moment bounds

$$p \in \mathbf{N}_{\geq 2}, \qquad \frac{\mathbf{E}\left[|X|^p\right]}{p!} \leq \frac{v}{2} \cdot b^{p-2},$$

then it follows that $X \in \beta(v,b)$. With this in mind, we will seek to compute bounds on $\mathbf{E}\left[|X|^p\right]$, taking a detour through the complex plane to do so.

Defining $A(t) = \mathbf{E}\left[\exp\left(t \cdot |X|\right) - 1 - t \cdot |X| - \frac{t^2}{2} \cdot |X|^2\right]$, observe quickly that A is well-defined on the real interval $|t| \leq h$. Moreover, defining $\xi(r) := \exp r - 1 - r$, observe that ξ is non-negative, that $\xi(r) \geq \xi(-r)$ for $r \geq 0$, and hence that

$$\xi\left(|r|\right) = \max\left\{\xi(r), \xi(-r)\right\} \leq \xi(r) + \xi(-r)$$

on all of $\mathbf{R}$. As a consequence, one can argue that for $0 \leq t \leq h$,

$$\begin{aligned} A(t) &= \mathbf{E}\left[\xi(t \cdot |X|) - \frac{t^2}{2} \cdot |X|^2\right] \\ &\leq \mathbf{E}\left[\xi(t \cdot X) + \xi(-t \cdot X) - \frac{t^2}{2} \cdot |X|^2\right] \\ &\leq 2 \cdot (M-1) - \frac{v}{2} \cdot t^2. \end{aligned}$$

Non-negativity of $|X|$ yields that $A(-t) \leq A(t)$ for $t \geq 0$, and convexity of A finally yields that for $|t| \leq h$, the maximal value of A is attained at the boundary (in particular, at the right boundary), so that there holds the bound $A(t) \leq 2 \cdot (M-1) - \frac{v}{2} \cdot h^2$.

Considering now complex t, one uses that $|\mathbf{E}[\exp(t \cdot |X|)]| \leq \mathbf{E}[\exp(\Re t \cdot |X|)]$ to see that the same bound holds on the entire strip $\{t \in \mathbf{C} : |\Re t| \leq h\}$, which certainly contains the circle $\{t \in \mathbf{C} : |t| = h\}$. By standard arguments, A is even analytic on this complex strip.

Now, recall that for $p \geq 3$, $A^{(p)}(0) = \mathbf{E}[|X|^p]$. Moreover, by Cauchy's integral formula, we can write

$$\frac{A^{(p)}(0)}{p!} \leq \frac{\sup_{\mathbf{C}}\{|A(t)| : |t| = h\}}{h^p} \leq \frac{2 \cdot (M-1) - \frac{v}{2} \cdot h^2}{h^p}$$

to quickly obtain that $\frac{\mathbf{E}[|X|^p]}{p!} \leq \frac{2 \cdot (M-1) - \frac{v}{2} \cdot h^2}{h^p}$. In order to deduce that $\frac{\mathbf{E}[|X|^p]}{p!} \leq \frac{v}{2} \cdot b^{p-2}$ for all $p \geq 3$, it is thus sufficient to check that

$$\forall p \in \mathbf{N}_{\geq 3}, \qquad \frac{2 \cdot (M-1) - \frac{v}{2} \cdot h^2}{h^p} \leq \frac{v}{2} \cdot b^{p-2} \qquad \Longleftrightarrow \qquad \frac{4 \cdot (M-1) - v \cdot h^2}{v \cdot h^2} \leq (b \cdot h)^{p-2}.$$

For this condition to hold, it is equivalent that

$$b \cdot h \geq \sup\left\{\left[\frac{4 \cdot (M-1) - v \cdot h^2}{v \cdot h^2}\right]^{\frac{1}{p-2}} : p \in \mathbf{N}_{\geq 3}\right\} = \max\left\{1, \frac{4 \cdot (M-1) - v \cdot h^2}{v \cdot h^2}\right\}.$$

Noting that

$$M \geq \frac{\mathbf{E}[\exp(h \cdot X)] + \mathbf{E}[\exp(-h \cdot X)]}{2} = \mathbf{E}[\cosh(h \cdot X)] \geq \mathbf{E}\left[1 + \frac{h^2}{2} \cdot |X|^2\right] = 1 + \frac{v}{2} \cdot h^2,$$

one sees quickly that the maximum is always saturated by the second term (if not always uniquely), and we conclude. □

Remark 2. For a faster and marginally looser bound, one can instead work with $\widetilde{A}(t) = \mathbf{E}[\exp(t \cdot |X|)]$ (which is plainly upper-bounded by $2 \cdot M$ for $\{t \in \mathbf{C} : |t| = h\}$ and carry out the same argument to obtain $b_\star = \frac{4 \cdot M}{v \cdot h^3}$. The earlier approach of Pinelis [4] sharpens this approach by a factor of $\sqrt{3} \geq 1.7$, using a closely-related real-analytic approach.

3. Bernstein Bound is a Compound Poisson-Exponential Comparison

While the sub-Gaussian family is directly obtained by comparison to the CGF of a Gaussian random variable, the sub-Gamma family is only spiritually related to the CGF of a Gaussian random variable. Additionally, to the author's knowledge, there is no canonical 'Bernstein random variable'. Nevertheless, the CGF bounds used in this context do indeed (almost) correspond to the CGF of a rather explicit random variable. In this section, we give a precise description of this random variable.

Proposition 3. *With $v \geq 0$, $b > 0$, let $N \sim \mathsf{Poisson}\left(\frac{v}{2 \cdot b^2}\right)$, let $\{Y_i : i \geq 0\} \overset{\text{iid}}{\sim} \mathsf{Exponential}\left(rate = b^{-1}\right)$, set*

$$Z = \begin{cases} 0 & N = 0 \\ \sum_{1 \leq i \leq N} Y_i & N > 0, \end{cases}$$

and set $X = Z - \frac{v}{2 \cdot b}$. It then holds for $t < b^{-1}$ that

$$\log(\mathbf{E}[\exp(t \cdot X)]) = \frac{v \cdot t^2}{2 \cdot (1 - b \cdot t)}.$$

Proof. Compute first that for $t < b^{-1}$, $\mathbf{E}\left[\exp\left(t \cdot Y_i\right)\right] = \left(1 - b \cdot t\right)^{-1}$, and that for $s \geq 0$, $\mathbf{E}\left[s^N\right] = \exp\left(\frac{v}{2 \cdot b^2} \cdot (s-1)\right)$. Conditioning, one then computes that

$$\begin{aligned}
\mathbf{E}\left[\exp\left(t \cdot Z\right)\right] &= \mathbf{E}\left[\mathbf{E}\left[\exp\left(t \cdot Z\right) \mid N\right]\right] \\
&= \mathbf{E}\left[\mathbf{E}\left[\exp\left(t \cdot Y_0\right)\right]^N\right] \\
&= \exp\left(\frac{v}{2 \cdot b^2} \cdot \left(\mathbf{E}\left[\exp\left(t \cdot Y_0\right)\right] - 1\right)\right) \\
&= \exp\left(\frac{v}{2 \cdot b^2} \cdot \left(\frac{1}{1 - b \cdot t} - 1\right)\right) \\
&= \exp\left(\frac{v \cdot t}{2 \cdot b} + \frac{v \cdot t^2}{2 \cdot (1 - b \cdot t)}\right).
\end{aligned}$$

Recalling that $X = Z - \frac{v}{2 \cdot b}$, the result then follows. □

4. Acknowledgments

A MathOverflow answer of Iosif Pinelis [4] initially yielded a version of Proposition 1 with the bound $b_\star \leq \frac{4 \cdot M}{\sqrt{3} \cdot v \cdot h^3}$, using only classical techniques of real analysis (i.e. without recourse to complex-analytic arguments). The suggestion to use a complex-analytic argument to control the absolute moments was provided by Gemini 3 Flash, with an initial proof which was slightly careless about the relationship between derivatives of the MGF and (absolute) moments; I then cleaned this up. The final step to show that one of the two terms in the maximum is irrelevant is also taken from Pinelis' proof. The representation of the Bernstein CGF as a Compound Poisson distribution was also provided by Gemini 3 Flash; this interaction also proposed some connections to Tweedie-type models, which I elected not to develop further in this note. All writing was performed by the author.

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